02-56000-24

Original Effective Date: 11/15/00

Reviewed: 05/28/26

Revised: 10/01/26

Subject: Infertility

THIS MEDICAL COVERAGE GUIDELINE IS NOT AN AUTHORIZATION, CERTIFICATION, EXPLANATION OF BENEFITS, OR A GUARANTEE OF PAYMENT, NOR DOES IT SUBSTITUTE FOR OR CONSTITUTE MEDICAL ADVICE. ALL MEDICAL DECISIONS ARE SOLELY THE RESPONSIBILITY OF THE PATIENT AND PHYSICIAN. BENEFITS ARE DETERMINED BY THE GROUP CONTRACT, MEMBER BENEFIT BOOKLET, AND/OR INDIVIDUAL SUBSCRIBER CERTIFICATE IN EFFECT AT THE TIME SERVICES WERE RENDERED. THIS MEDICAL COVERAGE GUIDELINE APPLIES TO ALL LINES OF BUSINESS UNLESS OTHERWISE NOTED IN THE PROGRAM EXCEPTIONS SECTION.

Position Statement	Billing/Coding	Reimbursement	Program Exceptions	Definitions	Related Guidelines
Other	References	Updates			

DESCRIPTION:

Infertility diagnosis, treatment, assisted reproductive technologies (e.g., IVF, ZIFT, GIFT) and artificial and intrauterine insemination (AI) vary by member's benefits.

Infertility is defined as inability to conceive a child despite trying for one year, or the inability of the woman to carry a pregnancy that results in live birth.

Some causes of infertility include the following:

- Abnormalities in the semen (male factor infertility)
- Ovulatory disorders (ovulatory factor)
- Tubal injury, blockage, paratubal adhesions, or endometriosis (tubal/peritoneal factor)
- Abnormalities in cervical mucus-sperm interaction (cervical factor)
- Rarer conditions such as uterine abnormalities, immunologic aberrations, and infections.

In some cases, no specific cause of infertility is detected despite an extensive and complete evaluation.

Assisted reproductive technologies (ART) refer to several interventions designed to establish a viable pregnancy for individuals who have been diagnosed with infertility; due either to female factors (e.g., pelvic adhesions, ovarian dysfunction, endometriosis, prior tubal ligation), male factors (e.g., abnormalities in sperm; production, function, transport, or prior vasectomy), a combination of both male and female factors, or unknown causes.

Assisted reproductive technologies include: in vitro fertilization (IVF), gamete intrafallopian transfer (GIFT), zygote intra-fallopian transfer (ZIFT), donor oocytes, and cryopreserved embryo transfers (CET). ART incorporates some type of in vitro fertilization (IVF) procedure in which oocytes harvested from the female are inseminated in vitro with sperm harvested from the male. Following the fertilization procedure, the zygote is cultured and ultimately transferred back into the female's uterus or fallopian tubes (zygote intrafallopian transfer [ZIFT]). In some instances, the oocyte and sperm are collected, but no in vitro fertilization takes place, and the gametes are reintroduced into the fallopian tubes (gamete intrafallopian transfer [GIFT]).

Artificial insemination (AI) encompasses a variety of procedures (e.g., intracervical, intrauterine (IUI)) involving placement of whole semen or processed sperm into the female reproductive tract, which permits sperm-oocyte interaction in the absence of intercourse. AI uses processed sperm from the male partner or a donor.

ART techniques include: intracytoplasmic sperm injection (ICSI), assisted hatching, co-culture of embryos, and cryopreservation of reproductive tissue (e.g., ovarian, oocytes, testicular).

Intracytoplasmic sperm injection (ICSI) is performed in cases of male factor infertility when either insufficient numbers of sperm, abnormal morphology, or poor motility preclude unassisted in vitro fertilization.

Assisted hatching in vitro fertilization is implantation of the embryo in the uterus. It is hypothesized that during the in vitro component of the IVF, the zona pellucid becomes hardened, thus impairing the hatching process. Alternatively, some embryos may have some inherent inability to induce thinning of the zona pellucida before hatching. Mechanical disruption of the zona pellucida (e.g., assisted hatching) has been proposed as a mechanism to improve implantation rates.

Embryo co-culture techniques represent an effort to improve the culture media for embryos such that a geater proportion of embryos will reach the blastocyst stage, in an attempt to improve implantation and pregnancy rates. A variety of co-culture techniques have been investigated, involving the use of feeder cell layers derived from a range of tissues, including the use of human reproductive tissues (e.g., oviducts) to non-human cells (e.g., fetal bovine uterine or oviduct cells) to established cell lines (e.g., Vero cells or bovine kidney cells). Co-culture techniques techniques have not been standaradized.

Cryopreservation of ovarian tissue with subsequent auto or heterotopic transplant has been investigated as a technique to sustain the reproductive function of women or children who are faced with sterilizing procedures, such as chemotherapy, radiation therapy, surgery, or due to malignant diseases. Cryopreservation of ovarian tissue techniques is not standardized.

Cryopreservation of oocytes has been examined as a fertility preservation option for reproductive-age women undergoing cancer treatment. The mature oocyte is very fragile due to its large size, high water content, and chromosomal arrangement. There are 2 primary approaches to cryopreservation: a controlled-rate, slow-cooling method and a flash-freezing process known as vitrification. Vitrification, the newer method, is faster and requires a higher concentration of cryoprotectants.

Cryopreservation of testicular tissue/testicular sperm extraction (TESE) refers to the collection of sperm from testicular tissue in men with azoospermia. TESE may be performed at the time of a diagnostic biopsy, or performed as a subsequent procedure, specifically for the collection of spermatozoa. The spermatozoa may be isolated immediately and a portion used for an ICSI procedure at the time of oocyte retrieval from the partner, with the remainder cryopreserved. Alternatively, the entire tissue sample can be cryopreserved with portion thawed and sperm isolation performed at subsequent ICSI cycles.

Cryopreservation of testicular tissue in prepubertal boys may be considered in those undergoing chemotherapy for cancer. The goal of cryopreservation in prepubertal boys undergoing chemotherapy is maintenance of fertility potential by auto-transplanting the testicular tissue when chemotherapy has been completed.

Blastocyst transfer refers to extending the duration of embryo culture to the blastocyst stage for assisted reproduction. Blastocyst transfer offers several advantages compared with cleavage-stage (day 2-3) embryo transfers. Advantages of blastocyst transfer include the following: a higher implantation rate and livebirth rates, the opportunity to potentially select the most viable embryo(s) for transfer, the potential decrease in the number of embryos transferred, and better temporal synchronization between embryo and endometrium at the time of embryo transfer since implantation in vivo generally occurs on day 5–7.

Summary and Analysis of Evidence: An UpToDate review on “In vitro fertilization: Overview of clinical issues and questions” (Ho) states that “In vitro fertilization (IVF) refers to a complex procedure designed to overcome infertility and produce a live birth as a direct result of the intervention; it is one type of assisted reproductive technology (ART). In general, IVF involves stimulating ovaries with a combination of fertility medications and retrieving oocyte(s) from ovarian follicles. The retrieved oocytes may be cryopreserved for future use or fertilized in the laboratory (i.e., in vitro) to create embryos. The resultant

embryo(s) are transferred into the uterine cavity. These steps typically occur over approximately a two-week interval of time, which is called an IVF cycle. ART includes "all interventions that include the in vitro handling of both human oocytes and sperm or of embryos for the purpose of reproduction. This includes, but is not limited to, in vitro fertilization (IVF) and embryo transfer (ET), intracytoplasmic sperm injection (ICSI), embryo biopsy, preimplantation genetic testing (PGT), assisted hatching, gamete intrafallopian transfer, zygote intrafallopian transfer, gamete and embryo cryopreservation, semen, oocyte and embryo donation, and gestational carrier cycles. In vitro fertilization (IVF) is "a sequence of procedures that involves extracorporeal fertilization of gametes. It includes conventional in vitro insemination and ICSI". IVF involves ovarian stimulation of the female to create oocytes, oocyte retrieval, in vitro insemination, and transfer of the resultant embryos to a uterus. Intracytoplasmic sperm injection (ICSI) is "a procedure in which a single spermatozoon is injected into the oocyte cytoplasm". This procedure is typically used for severe male factor infertility. Intrauterine insemination (IUI) describes "a procedure in which laboratory processed sperm are placed in the uterus to attempt a pregnancy". IVF, including intracytoplasmic sperm injection (ICSI), is selected for individuals for severe male factor infertility.

POSITION STATEMENT:

NOTE: Coverage for infertility is subject to the member's benefit terms, limitations and maximums. Refer to contract language regarding infertility.

If the benefit for coverage for infertility treatment is available and there is documentation of infertility for at least one year's duration, the following indications **meet the definition of medical necessity:**

- Diagnostic test or procedures to determine female infertility and or male infertility
- Artificial or intrauterine insemination (AI). **NOTE:** For AI, during one ovulatory cycle, sperm may be inseminated twice (once before human chorionic gonadotropin (hCG) and once after hCG. This is considered to be two attempts during one cycle)
- Assisted Reproductive Technologies (ARTs): IVF, ZIFT, GIFT
- Four (4) cycles for any Assisted Reproductive Technologies (ARTs): – IVF, ZIFT, GIFT, AI
- A cycle (ovulatory) of infertility treatment may include ovulation monitoring, egg retrieval, sperm retrieval, embryo transfer, egg fertilization. **NOTE:** If medical circumstances warrant consideration of more than four (4) cycles, additional cycles will be reviewed for medical appropriateness (e.g., a new cause of infertility is discovered and treated during covered therapy)
- Services associated with the diagnosis and treatment of infertility.

For fertility drug coverage, refer to member's contract benefits.

For diagnostic testing coverage for infertility, refer to member's contract benefits.

NOTE: Family balancing is not eligible for coverage. Family balancing is a measured approach to a non-medically indicated use of preconception gender selection. Family balancing provides couples having at least one child the opportunity to use MicroSort to increase the chance of having another child of the less represented sex in the family.

The following assisted reproductive techniques **meet the definition of medically necessity**:

- Cryopreservation of testicular tissue/sperm in adult men with azoospermia as part of an intracytoplasmic sperm injection (ICSI) procedure
- Intracytoplasmic sperm injection (ICSI) for male factor infertility
- Blastocyst transfer

The following assisted reproductive techniques are considered **experimental or investigational**.

Assisted hatching is considered **experimental or investigational**, as there is insufficient clinical evidence to support the use of assisted hatching. The evidence is insufficient to determine that this technology results in an improvement in health outcomes.

Embryo co-culture of oocyte(s)/embryos is considered **experimental or investigational**. The evidence is insufficient to determine that this technology results in an improvement in health outcomes.

Cryopreservation of ovarian tissue, including storage and thawing of ovarian tissue is considered **experimental or investigational**. The evidence is insufficient to determine that this technology results in an improvement in health outcomes.

Cryopreservation of oocyte(s), including storage and thawing of oocytes, is considered **experimental or investigational**. The evidence is insufficient to determine that this technology results in an improvement in health outcomes.

Cryopreservation of testicular tissue, including storage and thawing of testicular tissue in prepubertal boys is considered **experimental or investigational**. The evidence is insufficient to determine that this technology results in an improvement in health outcomes.

Intracytoplasmic sperm injection in the absence of male factor infertility is considered **experimental or investigational**. The evidence is insufficient to determine that this technology results in an improvement in health outcomes.

BILLING/CODING INFORMATION:

The following codes may be used to report services associated with infertility and diagnostic testing for infertility (not all inclusive).

NOTE: Coverage for infertility and diagnostic testing for infertility is subject to the member's benefit terms, limitations and maximums. Refer to contract language regarding infertility.

CPT Coding:

10021	Fine needle aspiration; without imaging guidance
10022	Fine needle aspiration; with imaging guidance
1062T	Artificial insemination, transcervical intratubal, unilateral or bilateral, including imaging guidance
54500	Biopsy of the testis, needle
54800	Biopsy of epididymis, needle
55400	Vasovasostomy, vas vasovasorrhaphy
55870	Electroejaculation (may be used in patients who are unable to produce a normal ejaculate due to spinal cord or other nervous system disorder i.e., diabetic neuropathy)
58321	Artificial insemination; intra-cervical
58322	Artificial insemination; intra-uterine
58323	Sperm washing for artificial insemination
58345	Transcervical introduction of fallopian tube catheter for diagnosis and/or re-establishing patency (any method), with or without hysterosalpingography
58970	Follicle puncture for oocyte retrieval, any method
58974	Embryo transfer, intrauterine
58976	Gamete, zygote or embryo intrafallopian transfer, any method
89250	Culture of oocyte(s)/embryo(s), less than 4 days
89251	Culture of oocyte(s)/embryo(s), less than 4 days; with co-culture of oocyte(s)/embryos (**investigational**)
89253	Assisted embryo hatching, micro techniques (any method) (**investigational**)
89254	Oocyte identification from follicular fluid
89255	Preparation of embryo for transfer (any method)
89257	Sperm identification from aspirate (other than seminal fluid)
89258	Cryopreservation; embryo(s)
89259	Cryopreservation sperm
89260	Sperm isolation; simple prep (e.g., sperm wash and swim-up) for insemination or diagnosis with semen analysis

89261	Sperm isolation; complex prep (e.g., Percoll gradient, albumin gradient) for insemination or diagnosis with semen analysis
89264	Sperm identification from testis tissue, fresh or cryopreserved
89268	Insemination of oocytes
89272	Extended culture oocyte(s)/embryo(s), 4 – 7 days
89280	Assisted oocyte fertilization, micro technique; less than or equal to 10 oocytes
89281	Assisted oocyte fertilization, micro technique; greater than 10 oocytes
89290	Biopsy, oocyte polar body or embryo blastomere, micro technique (for pre-implantation genetic diagnosis); less than or equal to 5 embryos
89291	Biopsy, oocyte polar body or embryo blastomere, micro technique (for pre-implantation genetic diagnosis); greater than 5 embryos (**non-covered**)
89300	Semen analysis; presence and/or motility of sperm including Huhner test (post coital)
89310	Semen analysis; motility and count (not including Huhner test)
89320	Semen analysis; volume, count, motility, and differential
89321	Semen analysis; sperm presence and motility of sperm, if performed
89322	Semen analysis; volume, count, motility, and differential using strict morphologic criteria (e.g., Kruger)
89325	Sperm antibodies
89329	Sperm evaluation; hamster penetration test
89330	Sperm evaluation; cervical mucus penetration test, with or without spinnbarkeit test
89331	Sperm evaluation, for retrograde ejaculation, urine (sperm concentration, motility, and morphology, as indicated)
89335	Cryopreservation, reproductive tissue, testicular
89337	Cryopreservation, mature oocyte(s) (**investigational**)
89342	Storage, (per year); embryo(s)
89343	Storage, (per year); sperm/semen
89344	Storage, (per year); reproductive tissue, testicular/ovarian (**investigational**)
89346	Storage, (per year); oocyte (**investigational**)
89352	Thawing of cryopreserved; embryo(s)
89353	Thawing of cryopreserved; sperm/semen, each aliquot
89354	Thawing of cryopreserved; reproductive tissue, testicular/ovarian (**investigational**)
89356	Thawing of cryopreserved; oocytes, each aliquot (**investigational**)

HCPCS Coding:

S3655	Antisperm antibodies test (immunobead)
S4011	In vitro fertilization; including but not limited to identification and incubation of mature oocytes, fertilization with sperm, incubation of embryo(s), and subsequent visualization for determination of development
S4013	Complete cycle, gamete intrafallopian transfer (GIFT), case rate
S4014	Complete cycle, zygote intrafallopian transfer (ZIFT), case rate
S4015	Complete in vitro fertilization cycle, not other wise specified case rate
S4016	Frozen in vitro fertilization cycle, case rate
S4017	Incomplete cycle, treatment canceled prior to stimulation, case rate
S4018	Frozen embryo transfer procedure canceled before transfer, case rate
S4020	In vitro fertilization procedure cancelled before aspiration, case rate
S4021	In vitro fertilization procedure cancelled after aspiration, case rate
S4022	Assisted oocyte fertilization, case rate
S4023	Donor egg cycle, incomplete, case rate
S4025	Donor services for in vitro fertilization (sperm or embryo), case rate
S4026	Procurement of donor sperm from sperm bank
S4027	Storage of previously frozen embryos
S4028	Microsurgical epididymal sperm aspiration (MESA)
S4030	Sperm procurement and cryopreservation services; initial visit

S4031	Sperm procurement and cryopreservation services; subsequent visit
S4035	Stimulated intrauterine insemination (IUI), case rate
S4037	Cryopreserved embryo transfer, case rate
S4040	Monitoring and storage of cryopreserved embryos, per 30 days
S4042	Management of ovulation induction (interpretation of diagnostic tests and studies, non-face-to-face medical management of the patient), per cycle

REIMBURSEMENT INFORMATION:

Refer to section entitled **POSITION STATEMENT** and **BILLING/CODING INFORMATION**.

PROGRAM EXCEPTIONS:

Federal Employee Program (FEP): Follow FEP guidelines.

State Account Organization (SAO): Follow SAO guidelines.

Medicare Advantage products: No National Coverage Determinations (NCD) and/or Local Coverage Determination (LCD) was found at the time of the last guideline reviewed date.

The following Medicare Benefit Policy Manual Chapter 15-Covered Medical and Other Health Services was reviewed on the last guideline review date: 20.1 Treatment for Infertility, located at cms.gov.

If this Medical Coverage Guideline contains a step therapy requirement, in compliance with Florida law 627.42393, members or providers may request a step therapy protocol exemption to this requirement if based on medical necessity. The process for requesting a protocol exemption can be found at Coverage Protocol Exemption Request

DEFINITIONS:

ART: Assisted Reproductive Technologies.

Artificial insemination (AI): placement of semen into the uterus with a syringe.

Assisted hatching: Assisted hatching (AH) is a procedure in which the zona pellucida (outer covering) of the embryo is partially opened, usually by application of an acid or laser, to facilitate embryo implantation and pregnancy. (ASRM, 2014)

Azoospermia: lack of live spermatozoa in the semen.

Cryopreservation: special freezing process.

Cryopreserved embryo transfer (CET): once embryos have been fertilized in the lab, they are frozen (reserved) and later transferred to the uterus.

Direct intra-peritoneal insemination (DIPI): process-attempting fertilization inside the body by placing a needle into the abdomen.

Donor egg (DE): an egg donated from a woman other than the patient.

Electroejaculation: this procedure assists men who are unable to ejaculate; electricity is sent through the pelvic area to assist in ejaculation.

Embryo: an egg fertilized with sperm.

Fallopian tubes: tubes connecting the uterus to the area around the ovaries. Eggs travel through the tubes to reach the uterus.

Fertilization: egg penetrated by sperm to form an embryo.

Gamete intrafallopian transfer (GIFT): a form of IVF; eggs are harvested from the ovary, loaded into a tube with sperm, and immediately placed into the fallopian tube with special scope for fertilization inside the body.

Gamete: a mature male or female reproductive cell (sperm or ovum).

Hysterosalpingography: x-ray test where dye is injected into the uterus and fallopian tubes, to see their structure.

In vitro fertilization (IVF): the egg is fertilized with sperm in a dish in a laboratory, rather than inside a woman's body. The resulting embryo is placed into the uterus later. One "cycle" of IVF includes using medicines to stimulate the ovaries to ovulate, "harvesting" the eggs with an instrument, attempting to fertilize the eggs with sperm in the lab, and placing any embryos into the uterus.

Intra-cytoplasmic sperm injection (ICSI sperm): ICSI can be used to treat male infertility disorders, such as low sperm count, low sperm motility, abnormally shaped sperm, or azoospermia (complete absence of sperm in the man's ejaculate). In a dish in the lab, sperm is injected into the egg to fertilize it, rather than letting it penetrate the egg naturally. Embryos are transferred to the uterus.

Intrauterine: inside the uterus.

IUI: intrauterine insemination. A procedure, which involves placing sperm inside a women's uterus to facilitate fertilization.

Laparoscopy: a small cut is made in the skin, and a scope with a light on the end is placed inside the abdomen.

Microfertilization (egg drilling or tweaking): this technique creates a cut in the egg, so that sperm can get in more easily. A "subzonal microinjection" places the sperm between the egg and it's outer covering, and a "partial zonal dissection" cuts the egg before it is incubated with sperm. In either case, embryos are transferred from the lab dish into the uterus.

Microsurgical epididymal sperm aspiration (MESA): in men who were born without a vas deferens, there is no way for sperm to reach the penis. This procedure places a needle into the testicle area to aspirate sperm for fertilization.

Natural oocyte retrieval (NORIVF): harvesting eggs from the ovary, when the egg has ovulated naturally.

Obstruction: blockage.

Oocyte: egg.

Ovulation: the monthly process of eggs released from the ovaries.

Sperm penetration assay: lab test designed to check the ability of sperm to enter eggs. Other terms for this test are the heterologous ovum penetration test, and hamster ovum test. A hamster egg (ovum) is used as the target, and sperm are placed in a dish with these eggs for several hours. The number of eggs with sperm penetration is counted, and the semen sample is rated as "fertile" or "non-fertile".

Spermatozoa: a mature male germ cell, the specific output of the testes. It is the generative element of the semen, which serves to fertilize the ovum, and contains the genetic information to be transmitted to the zygote by the male.

Vas deferens: tube connecting the testicles to the penis. Sperm travels along this path during ejaculation.

Voluntary sterilization: choosing to undergo a process that makes a person unable to reproduce (e.g., tubal ligation (tubes tied) in women or vasectomy in males.

Zygote intra-fallopian transfer (ZIFT): a form of IVF. Eggs are harvested and fertilized in a dish in the laboratory. The fertilized egg is placed inside the fallopian tube.

Zygote: a fertilized egg that has not yet divided.

RELATED GUIDELINES:

None applicable.

OTHER:

Coverage for office visits, and infertility diagnostic services (e.g., diagnostic procedures to determine the cause of infertility (diagnosis), and laboratory procedures) are covered services if the member has a benefit to cover these services.

Other names used to report infertility:

Assisted Oocyte Fertilization (AOF), formerly known as ICSI
Assisted Hatching
Assisted Oocyte Fertilization
Assisted Reproductive Technologies (ART)
Blastocyst Transfer
Cryopreserved Embryo Transfer (CET)
Electroejaculation
Gamete Intrafallopian Transfer (GIFT)
Intracytoplasmic Sperm Injection (ICSI)
In Vitro Fertilization (IVF)
Natural Oocyte Retrieval (NORIVF)
Zygote Intrafallopian Transfer (ZIFT)

REFERENCES:

1. American College of Obstetricians and Gynecologists Committee Opinion No. 584: oocyte cryopreservation. Obstetrics & Gynecology 2014 Jan;123(1):221-2, Reaffirmed 2016.
2. American Society for Reproductive Medicine, Report on Management of Obstructive Azoospermia, 2006.
3. American Urological Association (AUA), Inc.Infertility Report on Evaluation of the Azoospermic Male (An AUA Best Practice Policy and American Society of Reproductive Medicine (ASRM) Practice committee Report), 2001.
4. Bahadur G, Chatterjee R, Ralph D. Testicular Tissue Cryopreservation in Boys. Ethical and Legal Issues: Case Report. Human Reproduction 2000; 15: 14-16-20 20.
5. Blue Cross Blue Shield Association Evidence Positioning System®. 4.02.04 Reproductive Techniques, 09/25.
6. Chronopoulou E, Raperport C, Sfakianakis A, et al. Elective oocyte cryopreservation for age-related fertility decline. J Assist Reprod Genet. 2021 May;38(5):1177-1186.
7. D'Angelo A, Amso N, Embryo Freezing for Preventing Ovarian Hyperstimulation Syndrome, 02/27/02.
8. Donnez J, Dolmans MM, Dennylle D et al. Live birth After Orthotopic Transplantation of Cryopreserved Ovarian Tissue. Lancet 2004; 364(9443): 1405-1410.
9. Fritz R, Jindal S. Reproductive aging and elective fertility preservation. J Ovarian Res. 2018 Aug 11;11(1):66.
10. Glujovsky D. Cleavage stage versus blastocyst stage embryo transfer in assisted reproductive technology. Cochrane Database Syst Rev. 2016.
11. Goodwin PJ, Oktay KH, Lobo RA. Options for Preservation of Fertility in Women. New England Journal of Medicine 2005; 1418-1420 (Correspondence).

12. Ho J. In vitro fertilization: Overview of clinical issues and questions. In: UpToDate, Post TW (Ed), UpToDate, Waltham, MA. (Accessed on March 31, 2025.)

13. Kaur P, Swarankar ML, Maheshwari M et al. A comparative study between cleavage stage embryo transfer at day 3 and blastocyst stage transfer at day 5 in in-vitro fertilization/intra-cytoplasmic sperm injection on clinical pregnancy rates. Journal of Human Reproductive Sciences 2014 Jul;7(3):194-197.

14. Lee SJ, Schover LR, Patridge AH. American Society of Clinical Oncology Recommendations on Fertility Preservation in Cancer Patients. Journal of Clinical Oncology 2006; 24(18): 2917-1931.

15. Lobo RA. Current Concepts: Potential Options for Preservation of Fertility in Women. New England Journal of Medicine 2005; 64-72.

16. Meirow D, Levron J, Eldar-Geva T et al. Pregnancy After Transplantation of Cryopreserved Ovarian Tissue in a Patient with Ovarian Failure After Chemotherapy. New England Journal of Medicine 2005; 318-321 (Correspondence).

17. Myers ER, McCrory DC, Mills AA, Price TM, Swamy GK, Tantibhedhyangkul J, Wu JM, Matchar DB. Effectiveness of Assisted Reproductive Technology. Evidence Report/Technology Assessment No. 167 (Prepared by the Duke University Evidence-based Practice Center under Contract No. 290-02-0025.) AHRQ Publication No. 08-E012. Rockville, MD: Agency for Healthcare Research and Quality. May 2008.

18. Oatway C, Gunby J Daya S. Day Three Versus Day Two Embryo Transfer Following In Vitro Fertilization or Intracytoplasmic Sperm Injection. Cochrane Database of Systematic Reviews 2004, Issue 2.

19. Oktay K, Buyuk E, Veeck L et al. Embryo Development After Heterotopic Transplantation of Cryopreserved Ovarian Tissue. Lancet 2004; 363: 837-840.

20. Palermo G, Joris H, Devroey P. Pregnancies After Intracytoplasmic Injection of Single Spermatozoon Into an Oocyte. The Lancet 1992; 340: 17-18.

21. Pandian Z, Bhattacharya S, Vale L, Templeton A. In Vitro Fertilization for Unexplained Subfertility. Cochrane Database of Systematic Reviews 2005, Issue 2.

22. Porcu E, Fabbri R, Damiano G et al. Clinical Experience and Applications of Oocyte Cryopreservation. Molecular and Cellular Endocrinology 2000; 169: 33-37.

23. Practice Committee of the American Society for Reproductive Medicine; Practice Committee of the Society for Assisted Reproductive Technology. Blastocyst culture and transfer in clinically assisted reproduction: a committee opinion. Fertil Steril. 2018 Dec;110(7):1246-1252.

24. Practice Committee of the American Society for Reproductive Medicine and the Practice Committee for the Society for Assisted Reproductive Technologies. Guidance on the limits to the number of embryos to transfer: a committee opinion. Fertil Steril. 2021 Sep;116(3):651-654.

25. Practice Committees of the American Society for Reproductive Medicine and the Society for Assisted Reproductive Technology. Intracytoplasmic sperm injection (ICSI) for non-male factor indications: a committee opinion. Fertil Steril. 2020 Aug;114(2):239-24.

26. Practice Committee of the American Society for Reproductive Medicine and the Practice Committee of the Society for Assisted Reproductive Technology. Ovarian tissue and oocyte cryopreservation. Fertility and Sterility 2008; 90: S241-S246.

27. Practice Committee of the American Society for Reproductive Medicine and Practice Committee of the Society for Assisted Reproductive Technology. Role of assisted hatching in in vitro fertilization: a guideline. Fertility and Sterility. Aug 2014;102(2):348-351.

28. Practice Committee of the American Society for Reproductive Medicine in collaboration with the Society for Male Reproduction and Urology. The management of obstructive azoospermia: a committee opinion. Fertil Steril. 2019 May;111(5):873-880.

29. Seif MMW, Edi-Osagie ECO, Farquhar C, Hooper L, Blake D, McGinlay P. Assisted Hatching on Assisted Conception (IVF & ICSI). Cochrane Database of Systematic Reviews 2006, Issue 1.
30. The Ethics committee of the American Society for Reproductive Medicine-Fertility Preservation and Reproduction in Cancer Patients. Fertility and Sterility 2005; 83(6): 1622-1628.
31. The National Women's Health Information Center U.S. Department of Health & Human Services-Infertility, 05/06.
32. Toledo AA, Wright G, Jones AE et al. Blastocyst Transfer: A Useful Tool for Reduction of High-Order Multiple Gestations in a Human Assisted Reproduction Program. American Journal of Obstetrics and Gynecology 2000; 183(2): 377-382.

COMMITTEE APPROVAL:

This Medical Coverage Guideline (MCG) was approved by the BCBSF Medical Policy and Coverage Committee on 05/28/26.

GUIDELINE UPDATE INFORMATION:

11/15/00	Medical Coverage Guideline Revised.
01/01/02	Annual HCPCS coding update.
04/01/02	HCPCS changes.
08/15/02	Added CPT codes (89254, 89320, 89321, 89325, 89329, 89330).
01/01/03	Annual review and annual HCPCS coding update.
01/01/04	Annual HCPCS coding update: revise descriptors for the following: 89250, 89251, and 89258. Deleted the following: 89252 and 89256. Added the following: 89268, 89272, 89280, 89281, 89290, 89291, 89335, 89342, 89343, 89344, 89346, 89352, 89353, 89354, and 89396. Changed 89251 from non-covered to investigational (per BCBSA medical policy). Added 0058T and 0059T (considered investigational). Revised program exception for Medicare & More.
12/15/04	Added code 58323 and descriptor under billing/coding information-in vitro laboratory procedures. Deleted code 89523 and replace with 89253. Deleted covered codes for Medicare & More under program exceptions (Medicare Policy retired). Added code 89264 (non-covered) for Medicare & More under program exceptions. Deleted coverage statement for State group and Tropicana & Fowler White.
01/01/05	Annual HCPCS coding update: revised code 89346 descriptor. Added S4042.
05/15/05	Deleted ICD-9 diagnoses that support medical necessity.
09/15/05	Added S4018.
10/15/06	Updated DESCRIPTION section to include information regarding: intracytoplasmic sperm injection (ICSI), assisted hatching, embryo co-culture, cryopreservation of ovarian and testicular tissue and oocytes. Added coverage statement (investigational) regarding "storage and thawing" of ovarian tissue and oocytes and cryopreservation of testicular tissue to the WHEN SERVICES ARE NOT COVERED section. Revised CPT BILLING/CODING INFORMATION section. Deleted code 89399 (invalid code) and 55899 (unlisted procedure, male genital system). Added code 89335 to list of procedures not covered by Medicare. Added azoospermia and spermatozoa to the DEFINITIONS section. Updated references.
01/01/07	Annual HCPCS coding update: deleted S4036.
08/15/07	Annual review, coverage statements maintained, guideline reformatted, references updated.
01/01/08	Annual HCPCS coding update: revised 89320 and 89321 descriptor. Added 89322 and 89331.
01/01/09	Annual HCPCS coding update: deleted 0058T and 0059T. Updated references.
12/15/09	Annual review; no change in position statement. Updated references.
01/01/11	Annual HCPCS coding update; added 0058T and 0059T.

12/15/11	Annual review; revised description, member's benefit statement, added information regarding blastocyst transfer to description, updated position statement (added assisting hatching and blastocyst transfer) and updated billing/coding and references.
12/15/12	Annual review; added cryopreservation of testicular tissue in adult men with azoospermia as part of an intracytoplasmic sperm injection procedure, intracytoplasmic sperm injection, blastocyst transfer and updated references.
05/11/14	Revision: Program Exceptions section updated.
07/01/14	Quarterly HCPCS update; added 0357T.
01/01/15	Annual HCPCS code update. Deleted 0059T. Added 89337.
09/15/16	Revision: Updated description section. Added statement for diagnostic testing for infertility to position statement section. Added note for infertility and diagnostic testing for infertility to the billing/coding information section. Added "for male factor infertility" to intracytoplasmic sperm injection. Added "including storage and thawing" to cryopreservation of testicular tissue experimental or investigational statement. Updated references.
1015/18	Review; Added intracytoplasmic sperm injection in the absence of male factor infertility to experimental or investigational statement. Added "sperm" to cryopreservation of testicular tissue statement. Revised position statements. Updated references.
01/01/20	Annual HCPCS coding update: code 0357T deleted.
01/01/21	Annual HCPCS code update. Deleted 0058T.
04/15/21	Review; no change in position statement. Updated references.
06/15/23	Review; revised description and position statement. Updated references.
06/15/24	Review; no change in position statement. Updated references.
06/15/25	Review; no change in position statement. Updated references.
06/15/26	Review; no change in position statement. Updated references.
10/01/26	Quarterly CPT/HCPCS update. Added 1062T.